Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/19/2006
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Project Title:	Modeling Microgravity Non-Premixed Combustion Systems		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60208-3125	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	11/16/2000	**End Date:**	11/15/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	3
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2511		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The task involves identifying and analyzing relatively simple models of combustion of non-premixed system; diffusion flame sheets and edge-flames. In particular, the task is to describe the development of intrinsic instabilities such as (i) spontaneous flame oscillations, the nature of the oscillations and the mechanisms responsible for such behaviors and (ii) cellular structures and how, in particular, they are affected by the flow field and the reactants compositions. Another objective is to investigate the effects of thermal expansion on diffusion flame instabilities. The objective of this work is to provide theoretical support to microgravity experimental studies of non-premixed combustion (diffusion flames) in an attempt to develop rigorous understanding of the effects of gravity, or the lack of gravity, on these processes. More specifically, the objective is to determine the conditions for the onset of flame instabilities, and identify the responsible physical mechanisms that lead to these modes of burning. Many practical applications and most fires consist of diffusion flames. Understanding fires is particularly important to NASA's missions because of its concern with the safety		

	of future space explorations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although the current project focuses on configurations most relevant to microgravity studies, the results being fundamantal in nature are also relevant to to the understanding of the structure and dynamics of diffusion flames which are most relevant to many earth applications. Many practical applications of combustion, such as furnaces and diesel engines, involve diffusion flames and in most cases fires result from the combustion of non-premixed reactants. Better understanding leads to improvement in design, combustion efficiency and, indirectly, reduction of pollutants.
Task Progress:	1. A novel design was introduced for the realization of a planar unstrained diffusion flame. In the Porous Plug Counter Diffusion (PPCD) burner the fuel is supplied from the bottom of the chamber through a semi-permeable plate that maintains a uniform flow and prevents backward diffusion of products. The oxidizer is supplied through an array of hundreds of closely spaced hypodermic needles, allowing the products to escape vertically through the space between the needles. As a result, a uniform flow is maintained throughout the chamber except, perhaps, for a narrow region adjacent to the tip of the needles that does not exceed the size of one diameter. The realization of a planar unstrained diffusion flame is likely to increase the experimental data on diffusion flame instability and facilitate direct comparison with theory. 2. A comprehensive stability analysis of a planar diffusion flame was presented. It was shown that when instabilities exist they are necessarily associated with excess or deficiency in the available enthalpy at the reaction sheet. This occurs as a result of differential - nonunity Lewis numbers - and preferential - unequal Lewis numbers - diffusion, and at high flow rates, namely when the Damkohler number is sufficiently low. Diffusion flame instabilities, therefore, occur at near-extinction conditions. The various patterns that are likely to be observed at the onset of instability are summarized for a wide range of Lewis numbers and for various mixture strengths. 3. The general characterization described above remains qualitatively correct even when realistically accounting for density variations, an effect that has not been studied before. Thermal expansion was found to have a different influence on the various modes of instability, and its overall effect depends on the ratio of the stoichiometric and supply temperatures as well as on the temperature differential across the entire combustion layer. 4. The main consequence of non-unity Lewis numbers on the combustion field is that it yields temperature and concentration profiles that are non-similar. A volumetric heat loss term that affects the temperature but not the concentration fields also generates non-similar profiles that can promote diffusive-thermal instabilities. In the presence of gas-phase radiation, for example, it was found that flame oscillation can be triggered by appreciable heat losses even for unity Lewis numbers, and the instability is enhanced by heat losses when the conditions already favor oscillations.
Bibliography Type:	Description: (Last Updated: 04/27/2010)
Articles in Peer-reviewed Journals	Kurdyumov V, Matalon M. "Dynamics of an edge-flame in a mixing layer." Combustion and Flame 2004 Dec;139(4):329-39. http://dx.doi.org/10.1016/j.combustflame.2004.09.005 , Dec-2004
Articles in Peer-reviewed Journals	Lo Jacono D, Papas P, Matalon M, Monkewitz PA. "An experimental realization of an unstrained planar diffusion flame." Proceedings of the Combustion Institute, 2005 Jan;30(1):501-9. http://dx.doi.org/10.1016/j.proci.2004.07.011 , Jan-2005
Articles in Peer-reviewed Journals	Metzener P, Matalon M. "Diffusive-thermal instabilities of diffusion flames: onset of cells and oscillations." Combustion Theory and Modelling, 2006 Aug;10(4):701-25. http://dx.doi.org/10.1080/13647830600719894 , Aug-2006
Articles in Peer-reviewed Journals	Metzener P, Matalon M. "The effect of thermal expansion on the stability of diffusion flames." submitted for publication January 2006, Combustion Theory and Modelling. , Jan-2006
Papers from Meeting Proceedings	Kurdyumov V, Matalon M. "Stabilization and onset of oscillation of an edge-flame in the near-wake of a fuel injector." 31st International Symposium on Combustion, Heidelberg, Germany, August 6 - 11, 2006. Proceedings of the 31st International Symposium on Combustion, 2006. , Aug-2006